

University of Minnesota
School of Physics and Astronomy

2026 Fall Mathematics 8302
Differential Topology
Assignment Solution

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Question 1.7

Let N denote the **north pole** $(0, \dots, 0, 1) \in \mathbb{S}^n \subseteq \mathbb{R}^{n+1}$, and let S denote the **south pole** $(0, \dots, 0, -1)$. Define the **stereographic projection** $\sigma : \mathbb{S}^n \setminus \{N\} \rightarrow \mathbb{R}^n$ by

$$\sigma(x_1, \dots, x_{n+1}) = \frac{(x_1, \dots, x_n)}{1 - x_{n+1}}. \quad (1)$$

Let $\tilde{\sigma}(x) = -\sigma(-x)$ for $x \in \mathbb{S}^n \setminus \{S\}$.

- (a) For any $x \in \mathbb{S}^n \setminus \{N\}$, show that $\sigma(x) = u$, where $(u, 0)$ is the point where the line through N and x intersects the linear subspace where $x_{n+1} = 0$. Similarly, show that $\tilde{\sigma}(x)$ is the point where the line through S and x intersects the same subspace. (For this reason, $\tilde{\sigma}$ is also called **stereographic projection from the south pole**.)
- (b) Show that σ is bijective and

$$\sigma^{-1}(u_1, \dots, u_n) = \frac{(2u_1, \dots, 2u_n, |u|^2 - 1)}{|u|^2 + 1}. \quad (2)$$

- (c) Compute the transition map $\tilde{\sigma} \circ \sigma^{-1}$ and verify that the atlas consisting of the two charts $(\mathbb{S}^n \setminus \{N\}, \sigma)$ and $(\mathbb{S}^n \setminus \{S\}, \tilde{\sigma})$ defines a smooth structure on $\mathbb{S}^n$. (The coordinates defined by σ and $\tilde{\sigma}$ are called **stereographic coordinates**.)
- (d) Show that this smooth structure is the same as the one defined in Example 1.31.

Example 1.31 We put a smooth structure on $\mathbb{S}^n$ as follows. For each $i = 1, \dots, n+1$, let $(U_i^\pm, \phi_i^\pm)$ denote the graph coordinate charts, where

$$U_i^\pm = \{(x_1, \dots, x_{n+1}) \in \mathbb{S}^n : \pm x_i > 0\}, \quad \phi_i^\pm(x_1, \dots, x_{n+1}) = (x_1, \dots, \hat{x}_i, \dots, x_{n+1}), \quad (3)$$

where the hat indicates that the i -th coordinate is omitted.

For any distinct indices i, j , the transition map $\phi_i^\pm \circ (\phi_j^\pm)^{-1}$ is easily computed. In the case $i < j$, we get

$$\phi_i^\pm \circ (\phi_j^\pm)^{-1}(u_1, \dots, u_n) = (u_1, \dots, \hat{u}_i, \dots, \pm \sqrt{1 - |u|^2}, \dots, u_n) \quad (4)$$

with the $\pm \sqrt{1 - |u|^2}$ in the j -th position, and a similar formula holds for $i > j$. In the case $i = j$, an even simpler formula holds:

$$\phi_i^\pm \circ (\phi_i^\mp)^{-1}(u_1, \dots, u_n) = \text{ID}_{\mathbb{B}^n}. \quad (5)$$

Thus, the collection of charts $\{(U_i^\pm, \phi_i^\pm)\}$ is a smooth atlas, and so defines a smooth structure on $\mathbb{S}^n$.

Answer

(a)

Consider the line

$$\ell(t) = N + t(x - N) = t(x_1, \dots, x_n, x_{n+1} - 1) + (0, \dots, 0, 1) = (tx_1, \dots, tx_n, t(x_{n+1} - 1) + 1). \quad (6)$$

Obviously, $\ell(0) = N$ and $\ell(1) = x$. We want to find the point where this line intersects the hyperplane $x_{n+1} = 0$. Setting the last coordinate of $\ell(t)$ equal to zero, we have

$$t(x_{n+1} - 1) + 1 = 0 \implies t = \frac{1}{1 - x_{n+1}}. \quad (7)$$

Plugging this value of t into the first n coordinates of $\ell(t)$, we get

$$\ell\left(t = \frac{1}{1 - x_{n+1}}\right) = \frac{(x_1, \dots, x_n)}{1 - x_{n+1}}. \quad (8)$$

This is exactly the definition of $\sigma(x)$. A similar argument shows that $\tilde{\sigma}(x)$ is the point where the line through S and x intersects the hyperplane $x_{n+1} = 0$. More explicitly, consider the line

$$\tilde{\ell}(t) = S + t(x - S) = t(x_1, \dots, x_n, x_{n+1} + 1) + (0, \dots, 0, -1) = (tx_1, \dots, tx_n, t(x_{n+1} + 1) - 1). \quad (9)$$

Setting the last coordinate of $\tilde{\ell}(t)$ equal to zero, we have

$$t(x_{n+1} + 1) - 1 = 0 \implies t = \frac{1}{1 + x_{n+1}}. \quad (10)$$

Plugging this value of t into the first n coordinates of $\tilde{\ell}(t)$, we get

$$\tilde{\ell}\left(t = \frac{1}{1 + x_{n+1}}\right) = \frac{(x_1, \dots, x_n)}{1 + x_{n+1}}. \quad (11)$$

On the other hand, by the definition of $\tilde{\sigma}$, we have

$$\tilde{\sigma}(x) = -\sigma(-x) = -\frac{(-x_1, \dots, -x_n)}{1 - (-x_{n+1})} = \frac{(x_1, \dots, x_n)}{1 + x_{n+1}}. \quad (12)$$

These two expressions agree, so $\tilde{\sigma}(x)$ is exactly the point where the line through S and x intersects the hyperplane $x_{n+1} = 0$.

(b)

In order to show that σ is bijective, we will show that it is both injective and surjective. For injectivity,

suppose that $\sigma(x) = \sigma(y)$ for some $x, y \in \mathbb{S}^n \setminus \{N\}$. Then we have

$$\frac{(x_1, \dots, x_n)}{1 - x_{n+1}} = \frac{(y_1, \dots, y_n)}{1 - y_{n+1}}. \quad (13)$$

Hence, we have

$$x_i = \frac{1 - x_{n+1}}{1 - y_{n+1}} y_i \quad \text{for all } i = 1, \dots, n. \quad (14)$$

Consider the last coordinate of x and y . Since both x and y are on the unit sphere, we have

$$1 - x_{n+1}^2 = \sum_{i=1}^n x_i^2 = \sum_{i=1}^n \left(\frac{1 - x_{n+1}}{1 - y_{n+1}} y_i \right)^2 = \left(\frac{1 - x_{n+1}}{1 - y_{n+1}} \right)^2 \sum_{i=1}^n y_i^2 = \left(\frac{1 - x_{n+1}}{1 - y_{n+1}} \right)^2 (1 - y_{n+1}^2). \quad (15)$$

This implies that

$$1 - x_{n+1}^2 = \left(\frac{1 - x_{n+1}}{1 - y_{n+1}} \right)^2 (1 - y_{n+1}^2) \quad (16)$$

$$\implies (1 - x_{n+1})(1 + x_{n+1}) = \frac{(1 - x_{n+1})^2}{(1 - y_{n+1})^2} (1 - y_{n+1})(1 + y_{n+1}) \quad (17)$$

$$\implies (1 + x_{n+1})(1 - y_{n+1}) = (1 - x_{n+1})(1 + y_{n+1}) \quad (18)$$

$$\implies 1 + x_{n+1} - y_{n+1} - x_{n+1}y_{n+1} = 1 - x_{n+1} + y_{n+1} - x_{n+1}y_{n+1} \quad (19)$$

$$\implies 2x_{n+1} = 2y_{n+1} \implies x_{n+1} = y_{n+1}. \quad (20)$$

If $x_{n+1} = y_{n+1}$, then we have

$$x_i = \frac{1 - x_{n+1}}{1 - y_{n+1}} y_i = y_i \quad \text{for all } i = 1, \dots, n. \quad (21)$$

Thus, we have shown that $x = y$, and hence σ is injective.

Next, in order to show that σ is surjective, let $u \in \mathbb{R}^n$ be arbitrary. We want to find $x \in \mathbb{S}^n \setminus \{N\}$ such that $\sigma(x) = u$. Let $x = (x_1, \dots, x_n, x_{n+1})$. Then we have

$$\sigma(x) = u \implies \frac{(x_1, \dots, x_n)}{1 - x_{n+1}} = u \implies (x_1, \dots, x_n) = (1 - x_{n+1})u = (u_1(1 - x_{n+1}), \dots, u_n(1 - x_{n+1})). \quad (22)$$

With the constraint that $x \in \mathbb{S}^n$, we have

$$1 = x_1^2 + \dots + x_n^2 + x_{n+1}^2 = (1 - x_{n+1})^2 |u|^2 + x_{n+1}^2 \implies 1 = (1 - x_{n+1})^2 |u|^2 + x_{n+1}^2. \quad (23)$$

This is a quadratic equation in x_{n+1} , which we can simplify as follows:

$$0 = (1 - x_{n+1})^2 |u|^2 + x_{n+1}^2 - 1 = (|u|^2 + 1)x_{n+1}^2 - 2|u|^2 x_{n+1} + |u|^2 - 1. \quad (24)$$

Solving for x_{n+1} , we have

$$x_{n+1} = \frac{2|u|^2 \pm \sqrt{4|u|^4 - 4(|u|^2 + 1)(|u|^2 - 1)}}{2(|u|^2 + 1)} = \frac{|u|^2 \pm 1}{|u|^2 + 1}. \quad (25)$$

Since we want $x \in \mathbb{S}^n \setminus \{N\}$, we must have $x_{n+1} \neq 1$, which means that we must take the negative sign in the above expression. Hence, we have

$$x_{n+1} = \frac{|u|^2 - 1}{|u|^2 + 1}. \quad (26)$$

Now that we have x_{n+1} , we can find x_i for $i = 1, \dots, n$:

$$x_i = u_i(1 - x_{n+1}) = u_i \left(1 - \frac{|u|^2 - 1}{|u|^2 + 1} \right) = u_i \left(\frac{2}{|u|^2 + 1} \right) = \frac{2u_i}{|u|^2 + 1}. \quad (27)$$

It remains to check that this x indeed has the required properties. First, we have

$$|x|^2 = \frac{4|u|^2 + (|u|^2 - 1)^2}{(|u|^2 + 1)^2} = \frac{|u|^4 + 2|u|^2 + 1}{(|u|^2 + 1)^2} = \frac{(|u|^2 + 1)^2}{(|u|^2 + 1)^2} = 1, \quad (28)$$

so $x \in \mathbb{S}^n$. Moreover, $1 - x_{n+1} = \frac{2}{|u|^2 + 1} \neq 0$, so $x \neq N$, and

$$\sigma(x) = \frac{(x_1, \dots, x_n)}{1 - x_{n+1}} = \frac{\frac{2}{|u|^2 + 1}(u_1, \dots, u_n)}{\frac{2}{|u|^2 + 1}} = u. \quad (29)$$

Thus, we have found $x \in \mathbb{S}^n \setminus \{N\}$ such that $\sigma(x) = u$, and hence σ is surjective. Therefore, σ is bijective. Besides, we have shown that

$$\sigma^{-1}(u_1, \dots, u_n) = \frac{(2u_1, \dots, 2u_n, |u|^2 - 1)}{|u|^2 + 1}. \quad (30)$$

(c)

The transition map $\tilde{\sigma} \circ \sigma^{-1}$ is defined on the image under σ of the overlap of the two charts. Since $\sigma^{-1}(u) = S$ if and only if $u = 0$, this image is

$$\sigma(\mathbb{S}^n \setminus \{N, S\}) = \mathbb{R}^n \setminus \{0\}. \quad (31)$$

For any $u \in \mathbb{R}^n \setminus \{0\}$, using the formula $\tilde{\sigma}(x) = \frac{(x_1, \dots, x_n)}{1 + x_{n+1}}$ obtained in part (a), we have

$$\tilde{\sigma} \circ \sigma^{-1}(u) = \tilde{\sigma} \left(\frac{(2u_1, \dots, 2u_n, |u|^2 - 1)}{|u|^2 + 1} \right) = \frac{\frac{1}{|u|^2 + 1}(2u_1, \dots, 2u_n)}{1 + \frac{|u|^2 - 1}{|u|^2 + 1}} = \frac{(2u_1, \dots, 2u_n)}{2|u|^2} \quad (32)$$

$$= \frac{(u_1, \dots, u_n)}{|u|^2}. \quad (33)$$

Every component of this map is a rational function of u whose denominator $|u|^2$ does not vanish on $\mathbb{R}^n \setminus \{0\}$, and hence $\tilde{\sigma} \circ \sigma^{-1}$ is smooth.

Besides, we need the other transition map $\sigma \circ \tilde{\sigma}^{-1}$ to be smooth as well. Write $f(u) = \frac{u}{|u|^2}$ for the map computed above. Then $|f(u)| = \frac{1}{|u|}$, and hence

$$f(f(u)) = \frac{f(u)}{|f(u)|^2} = \frac{u/|u|^2}{1/|u|^2} = u. \quad (34)$$

Thus f is its own inverse, so

$$\sigma \circ \tilde{\sigma}^{-1} = (\tilde{\sigma} \circ \sigma^{-1})^{-1} = f^{-1} = f \quad (35)$$

is given by the same formula, and is therefore smooth on $\mathbb{R}^n \setminus \{0\}$ as well.

Since $\mathbb{S}^n = (\mathbb{S}^n \setminus \{N\}) \cup (\mathbb{S}^n \setminus \{S\})$ and both transition maps are smooth, the atlas consisting of the two charts $(\mathbb{S}^n \setminus \{N\}, \sigma)$ and $(\mathbb{S}^n \setminus \{S\}, \tilde{\sigma})$ is a smooth atlas, and so defines a smooth structure on $\mathbb{S}^n$.

(d)

First, we consider the transition map $\phi_i^\pm \circ \sigma^{-1}$. Since $x_i = \frac{2u_i}{|u|^2+1}$ and $|u|^2 + 1 > 0$, the condition $\pm x_i > 0$ is equivalent to $\pm u_i > 0$ for $i = 1, \dots, n$, and the condition $\pm x_{n+1} > 0$ is equivalent to $\pm(|u|^2 - 1) > 0$. In either case the domain

$$\sigma(U_i^\pm \cap (\mathbb{S}^n \setminus \{N\})) \quad (36)$$

is an open subset of $\mathbb{R}^n$. For $i = 1, \dots, n$ and u in this set, we have

$$\phi_i^\pm \circ \sigma^{-1}(u) = \phi_i^\pm \left(\frac{(2u_1, \dots, 2u_n, |u|^2 - 1)}{|u|^2 + 1} \right) \quad (37)$$

$$= \left(\frac{2u_1}{|u|^2 + 1}, \dots, \frac{2u_{i-1}}{|u|^2 + 1}, \frac{2u_{i+1}}{|u|^2 + 1}, \dots, \frac{2u_n}{|u|^2 + 1}, \frac{|u|^2 - 1}{|u|^2 + 1} \right), \quad (38)$$

where the i -th coordinate has been omitted. For $i = n + 1$, we have

$$\phi_{n+1}^\pm \circ \sigma^{-1}(u) = \phi_{n+1}^\pm \left(\frac{(2u_1, \dots, 2u_n, |u|^2 - 1)}{|u|^2 + 1} \right) = \left(\frac{2u_1}{|u|^2 + 1}, \dots, \frac{2u_n}{|u|^2 + 1} \right). \quad (39)$$

In both cases every component is a rational function of u whose denominator $|u|^2 + 1$ is strictly positive, and hence the transition map is smooth.

Next, we consider the transition map $\sigma \circ (\phi_i^\pm)^{-1}$. On $U_i^\pm$ we have $x_i^2 = 1 - |v|^2 > 0$, so the image $\phi_i^\pm(U_i^\pm)$ is contained in the open unit ball $\mathbb{B}^n$, on which $\sqrt{1 - |v|^2}$ is smooth. For $i = 1, \dots, n$ the north

pole N does not lie in $U_i^\pm$, since its i -th coordinate is 0; hence the domain is all of $\mathbb{B}^n$, and

$$\sigma \circ (\phi_i^\pm)^{-1}(v) = \sigma \left((v_1, \dots, v_{i-1}, \underbrace{\pm \sqrt{1 - |v|^2}}_{\text{the } i\text{-th coordinate}}, v_i, \dots, v_n) \right) \quad (40)$$

$$= \frac{1}{1 - v_n} (v_1, \dots, v_{i-1}, \pm \sqrt{1 - |v|^2}, v_i, \dots, v_n). \quad (41)$$

Since $|v| < 1$ forces $|v_n| < 1$, the denominator $1 - v_n$ never vanishes, and hence this map is smooth. For $i = n + 1$, we have

$$\sigma \circ (\phi_{n+1}^\pm)^{-1}(v) = \sigma \left((v_1, \dots, v_n, \underbrace{\pm \sqrt{1 - |v|^2}}_{\text{the } (n+1)\text{-th coordinate}}) \right) = \frac{1}{1 \mp \sqrt{1 - |v|^2}} (v_1, \dots, v_n). \quad (42)$$

Here $N = (0, \dots, 0, 1) \in U_{n+1}^+$ and $\phi_{n+1}^+(N) = 0$, so for the $+$ sign the domain is the punctured ball

$$\phi_{n+1}^+(U_{n+1}^+ \cap (\mathbb{S}^n \setminus \{N\})) = \mathbb{B}^n \setminus \{0\}, \quad (43)$$

on which $1 - \sqrt{1 - |v|^2} > 0$. For the $-$ sign the domain is all of $\mathbb{B}^n$, and $1 + \sqrt{1 - |v|^2} \geq 1 > 0$. In both cases the denominator does not vanish, and hence the transition map is smooth.

Last, from $\tilde{\sigma}(x) = -\sigma(-x)$ we obtain $\tilde{\sigma}^{-1}(u) = -\sigma^{-1}(-u)$, that is,

$$\tilde{\sigma}^{-1}(u_1, \dots, u_n) = \frac{(2u_1, \dots, 2u_n, 1 - |u|^2)}{|u|^2 + 1}, \quad (44)$$

which differs from σ^{-1} only in the sign of the last coordinate. Likewise $\tilde{\sigma}(x) = \frac{(x_1, \dots, x_n)}{1 + x_{n+1}}$ differs from σ only in the sign appearing in the denominator. Hence the computations above go through verbatim, with the roles of the $+$ and $-$ signs interchanged in the case $i = n + 1$, since it is now $S = (0, \dots, 0, -1) \in U_{n+1}^-$ that has to be removed. Therefore both $\phi_i^\pm \circ \tilde{\sigma}^{-1}$ and $\tilde{\sigma} \circ (\phi_i^\pm)^{-1}$ are smooth.

Therefore, the smooth structure defined by stereographic coordinates is the same as the one defined in Example 1.31.

Question 1.9

Complex projective n -space, denoted by $\mathbb{CP}^n$, is the set of all 1-dimensional complex-linear subspaces of $\mathbb{C}^{n+1}$, with the quotient topology inherited from the natural projection $\pi : \mathbb{C}^{n+1} \setminus \{0\} \rightarrow \mathbb{CP}^n$. Show that $\mathbb{CP}^n$ is a compact $2n$ -dimensional topological manifold, and show how to give it a smooth structure analogous to the one we constructed for $\mathbb{RP}^n$. (We use the correspondence

$$(x^1 + iy^1, \dots, x^{n+1} + iy^{n+1}) \longleftrightarrow (x^1, \dots, x^{n+1}, y^1, \dots, y^{n+1}) \quad (45)$$

to identify $\mathbb{C}^{n+1}$ with $\mathbb{R}^{2n+2}$.)

Answer